

as: 15

Implement a YOLO model to detect objects.

Aim:

To implement a YOLO model to detect objects.

Objective:

→ To study how YOLO detects multiple objects in a single image.

→ To understand the architecture and working of a pre-trained YOLO model.

→ To use transfer learning for custom object detection.

Pseudo code:

1. Install and import (YOLO package)

2. Load a pre-trained YOLO model

3. Load a test image or use a camera frame

4. Run the model predict() method to detect objects.

5. Display bounding boxes and class labels.

6. Save the annotated output image.

webcam

object detection
method

XNOR-YOLO

prediction

Bounding Box

class 1 Ball at Goal

Ball and goal
coordinate

Robot will
take ball
and goal

Observations:

1. The pre-trained YOLO v3s model successfully detected multiple objects such as cars, buses, person in single frame.
2. Each object was enclosed in a bounding box with a clear label and confidence score.
3. The inference time per image was very low.
4. The visualization clearly showed YOLO's ability to detect overlapping objects in complex scenes.



Results:

successfully implemented pre-trained YOLO.

image 1(1) context 1img - 20250523 WA0017
640x480

Person - 7.3ms

speed : 3 preprocess : 7.3ms inference : 24ms

preprocess per image A shape

(1, 3, 640, 480)

output:

Detected Object

person

person

person

~~Can only support (H264, H265)
in a pre-trained YOLO model~~

It is a fast image or video camera frame
in the model predict () method to detect
objects.
In the following function boxes are drawn on
the image.